

Practice Problems-Sequential Circuits (Part 2)

Problem 1: A new flip-flop is having behaviour as described below. It has two inputs X and Y and when both inputs are same and they are 1,1 the flip-flop is going to set else flip-flop resets. If both inputs are different and they are 0,1 the flip-flop complements itself otherwise it is going to retain the last state. Which of the following expression is the characteristic expression for the new flip-flop?

a) $XQ + YQ'$

b) $XQ' + YQ$

c) $XQ' + YQ'$

d) None

Problem 2: Minimum

a) 3

c) 5

Sol: -



a) $XQ + YQ'$

b) $XQ' + YQ$

c) $XQ' + YQ'$

d) None

Sol:-

meb

Q	X	Y	Q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

New flip-flop

X Y

0 0 (reset, 0/p=0)

1 1 (set, 0/p=1)

0 1 complement of P.S.

1 0 memory

Q

X Y	00	01	11	10
0	0	1	1	0
1	0	0	1	1

Sol:-

Q	X	Y	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

0 0 (reset, 0/p=0)
1 1 (set, 0/p=1)

0 1 complement of P.S.
1 0 memory

for Q^+ :-

Q	XY	00	01	11	10
0		0	1	1	0
1		0	0	1	1

$$Q^+ = I + II$$

$$Q^+ = \bar{Q}Y + QX$$

Problem 2: Minimum number of flip-flops required in modulo-8 counter:

a) 3

b) 4



c) 5

d) 8

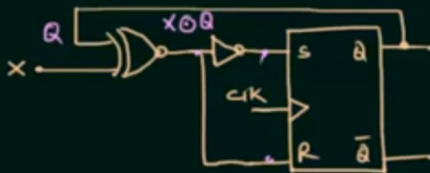
= 0)

)

P.S.

Practice Problems-Sequential Circuits (Part 3)

Problem 1: Consider the circuit shown in the figure:



The expression for the next state Q^+ is

a) XQ

b) XQ'

c) $X \oplus Q$

d) $X \odot Q$

$$0/P = Q^+ (N.S.)$$

$$1/P = X \odot Q (P.S.)$$

$$S = \overline{X \odot Q}$$

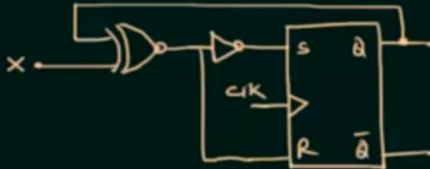
$$R = X \odot Q$$

X	Q	Q^+	S	R
0	0	0	0	1
0	1	1	1	0
1	0	1	1	0
1	1	0	0	1

$$Q^+ = X \oplus Q$$

Practice Problems-Sequential Circuits (Part 3)

Problem 1: Consider the circuit shown in the figure:



The expression for the next state Q^+ is

- a) XQ
- b) XQ'
- c) $X \oplus Q$
- d) $X \odot Q$

Problem 2: Match the column:

List I

a. Shift register

b. Counter

c. Decoder

List II

1. Frequency division

2. Addressing in memory chips

3. Serial to parallel data conversion